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E/21/245

Lab 02 - CO226 - Data Base

1. Recreate the following tables with relationships using MySQL syntaxes.

Publisher Table

Field Name	Type	Field Size	Constraints
PubID	Text (varchar)	6	Primary Key
PublisherName	Text (varchar)	30	
RegisteredDate	Date		
Country	Text (varchar)	15	

Book Table

Field Name	Type	Field Size	Constraints
BookID	Text (varchar)	5	Primary Key
Title	Text (varchar)	50	
Author	Text (varchar)	30	
Category	Number (int)		
Price	Currency (float)		
PubID	Text (varchar)	6	Foreign Key

1.

Input

Run SQL

```sql
CREATE TABLE Publisher (
    PubID VARCHAR(6) PRIMARY KEY,
    PublisherName VARCHAR(30),
    RegisteredDate DATE,
    Country VARCHAR(15)
);

-- Create the Book table with a foreign key referencing Publisher
CREATE TABLE Book (
    BookID VARCHAR(5) PRIMARY KEY,
    Title VARCHAR(50),
    Author VARCHAR(30),
    Category INT,
    Price FLOAT,
    PubID VARCHAR(6),
    FOREIGN KEY (PubID) REFERENCES Publisher(PubID)
```

Available Tables

Book

BookID	Title	Author	Category	Price	PubID
empty					

Publisher

PubID	PublisherName	RegisteredDate	Country
empty			

2. Insert following records to Publisher table and Book table.

Book

Book ID	Title	Author	Category	Price	PubID
B1	Excel	R. David	2	18.00	Pub02
B2	Computers	M. Thomas	4	25.00	Pub02
B3	Access	P. Paul	2	30.00	Pub03
B4	Arts	W. Shiva	1	10.00	Pub04
B5	Science	A. Rahul	3	9.00	Pub01

Publisher

PubID	PublisherName	RegisteredDate	Country
Pub01	A.Press	2001-05-12	Sri Lanka
Pub02	A.Books	2010-03-20	India
Pub03	K.Brill	2014-01-31	Poland
Pub04	A.Press	2015-02-13	Sri Lanka

2.

Input

```
INSERT INTO Book (BookID, Title, Author, Category, Price, PubID)
VALUES
('B1', 'Excel', 'R. David', 2, 18.00, 'Pub02'),
('B2', 'Computers', 'M. Thomas', 4, 25.00, 'Pub02'),
('B3', 'Access', 'P. Paul', 2, 30.00, 'Pub03'),
('B4', 'Arts', 'W. Shiva', 1, 10.00, 'Pub04'),
('B5', 'Science', 'A. Rahul', 3, 9.00, 'Pub01');
```

Input

```sql
INSERT INTO Publisher (PubID, PublisherName, RegisteredDate, Country)
VALUES
('Pub01', 'A.Press', '2001-05-12', 'Sri Lanka'),
('Pub02', 'A.Books', '2010-03-20', 'India'),
('Pub03', 'K.Brill', '2014-01-31', 'Poland'),
('Pub04', 'A.Press', '2015-02-13', 'Sri Lanka');
```

Available Tables

Book

BookID	Title	Author	Category	Price	PubID
B1	Excel	R. David	2	18	Pub02
B2	Computers	M. Thomas	4	25	Pub02
B3	Access	P. Paul	2	30	Pub03
B4	Arts	W. Shiva	1	10	Pub04
B5	Science	A. Rahul	3	9	Pub01

Publisher

PubID	PublisherName	RegisteredDate	Country
Pub01	A.Press	2001-05-12	Sri Lanka
Pub02	A.Books	2010-03-20	India
Pub03	K.Brill	2014-01-31	Poland
Pub04	A.Press	2015-02-13	Sri Lanka

3. Write SQL queries/statements considering the following table. Table Name: Book
   a. Display all the details in the Book table
   b. Display all the titles of the books.
   c. Display all the titles of the books with prices
   d. Display the Book ID, the Title and the Author.
   e. Display the details of the book with the Book ID B2.
   f. Display the titles of the books with the prices less than 20.00
   g. Display the details of the books in category 2.

a)

Input

Run SQL

```
SELECT * FROM Book
```

Output

BookID	Title	Author	Category	Price	PubID
B1	Excel	R. David	2	18	Pub02
B2	Computers	M. Thomas	4	25	Pub02
B3	Access	P. Paul	2	30	Pub03
B4	Arts	W. Shiva	1	10	Pub04
B5	Science	A. Rahul	3	9	Pub01

b)

Input

Run SQL

```
SELECT Title FROM Book
```

Output

Title
Excel
Computers
Access
Arts
Science

c)

Input

Run SQL

```
SELECT Title,Price FROM Book
```

Output

Title	Price
Excel	18
Computers	25
Access	30
Arts	10
Science	9

d)

```sql
SELECT BookID,Title,Author FROM Book
```

Output

BookID	Title	Author
B1	Excel	R. David
B2	Computers	M. Thomas
B3	Access	P. Paul
B4	Arts	W. Shiva
B5	Science	A. Rahul

e)

Input Run SQL

```sql
SELECT * FROM Book
WHERE BookID = "B2"
```

Output

BookID	Title	Author	Category	Price	PubID
B2	Computers	M. Thomas	4	25	Pub02

f)

Input

Run SQL

```
SELECT Title FROM Book
WHERE Price < 20.00
```

Output

Title
Excel
Arts
Science

g)

Input

Run SQL

```
SELECT * FROM Book
WHERE Category=2
```

Output

BookID	Title	Author	Category	Price	PubID
B1	Excel	R. David	2	18	Pub02
B3	Access	P. Paul	2	30	Pub03

4. Write queries
   a. Display all the Titles in the Book table in ascending order.
   b. Display all the details in the Book table in descending order by Price.
   c. Display all the details in the Book table in ascending order by Price and Title.

a)

Input

Run SQL

```
SELECT Title FROM Book ORDER BY Title ASC;
```

Output

Title
Access
Arts
Computers
Excel
Science

b)

Input

Run SQL

```
SELECT * FROM Book ORDER BY Price DESC;
```

Output

BookID	Title	Author	Category	Price	PubID
B3	Access	P. Paul	2	30	Pub03
B2	Computers	M. Thomas	4	25	Pub02
B1	Excel	R. David	2	18	Pub02
B4	Arts	W. Shiva	1	10	Pub04
B5	Science	A. Rahul	3	9	Pub01

c)

Input

Run SQL

```
SELECT * FROM Book ORDER BY Price ASC, Title ASC;
```

Output

BookID	Title	Author	Category	Price	PubID
B5	Science	A. Rahul	3	9	Pub01
B4	Arts	W. Shiva	1	10	Pub04
B1	Excel	R. David	2	18	Pub02
B2	Computers	M. Thomas	4	25	Pub02
B3	Access	P. Paul	2	30	Pub03

5. Write SQL queries/statements considering the ‘Book’ table
    a. Change the category as 3 of the book with the BookID ‘B3’.
    b. Change the price of the book with the BookID ‘B3’ by adding 10.00 to the current price.
    c. Delete the details of the book with the BookID ‘B3’.
    d. Delete all the books in category 2.
    e. Remove all the data in the Book table.

a)

Input

Run SQL

Available Tables

```
UPDATE Book SET Category = 3 WHERE BookID = "B3";
```

Book

BookID	Title	Author	Category	Price	PubID
B1	Excel	R. David	2	18	Pub02
B2	Computers	M. Thomas	4	25	Pub02
B3	Access	P. Paul	3	30	Pub03
B4	Arts	W. Shiva	1	10	Pub04
B5	Science	A. Rahul	3	9	Pub01

b)

Input

Run SQL

Available Tables

```
UPDATE Book SET Price = Price + 10.00 WHERE BookID = "B3";
```

Book

BookID	Title	Author	Category	Price	PubID
B1	Excel	R. David	2	18	Pub02
B2	Computers	M. Thomas	4	25	Pub02
B3	Access	P. Paul	3	40	Pub03
B4	Arts	W. Shiva	1	10	Pub04
B5	Science	A. Rahul	3	9	Pub01

c)

Input

Run SQL

Available Tables

```
DELETE FROM Book WHERE BookID = 'B3';
```

Book

BookID	Title	Author	Category	Price	PubID
B1	Excel	R. David	2	18	Pub02
B2	Computers	M. Thomas	4	25	Pub02
B4	Arts	W. Shiva	1	10	Pub04
B5	Science	A. Rahul	3	9	Pub01

d)

Input

Run SQL

Available Tables

```
DELETE FROM Book WHERE Category = 2;
```

Book

BookID	Title	Author	Category	Price	PubID
B2	Computers	M. Thomas	4	25	Pub02
B4	Arts	W. Shiva	1	10	Pub04
B5	Science	A. Rahul	3	9	Pub01

e)

Input

Run SQL

Available Tables

```
DELETE FROM Book;
```

Book

BookID	Title	Author	Category	Price	PubID
empty					